

Grokking Larger DLP with Fixed Model Size: $O(1)$ Representations for Scalable Discrete Logarithm Learning

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Abstract

We study whether the delayed generalization phenomenon (“grokking”) on the finite-field discrete logarithm problem (DLP) can scale to large primes without increasing model size. Our companion paper [Svensson, 2026a] demonstrated DLP grokking on small primes ($p \leq 521$) but identified a fundamental “tokenizer confound”: both the input embedding and output classifier grow linearly with p , so model parameter count scales as $O(p)$. Here we replace the atomic tokenizer with two $O(1)$ components—Factorized Residue Embedding (FRE) for inputs and Factorized Candidate Output (FCO) for outputs—combined with a CRT-BOTH representation that decomposes the DLP into two coprime subproblems via the Chinese Remainder Theorem. The resulting architecture maintains a fixed $\sim 530\text{k}$ parameters regardless of prime size. We demonstrate that this fixed-size model groks DLP across a $35\times$ range of prime sizes ($p = 113$ to $p = 4001$), achieving 100% test accuracy at every scale. The parameter efficiency advantage grows with p : at $p = 16411$ the fixed model is $8.7\times$ smaller than the atomic baseline, and at $p = 300,017$ it would be $56\times$ smaller. We present a scaling plan extending to $p = 300,017$ and describe fallback strategies ($2\times$ model size, $2\times$ dataset size) for primes where the base configuration fails to grok. Our results suggest that representation choice, not model capacity, is the primary bottleneck for scaling grokking to large algebraic problems.

1 Introduction

The *grokking* phenomenon [Power et al., 2022]—in which a neural network abruptly transitions from memorization to generalization long after overfitting—has been studied extensively on modular arithmetic [Nanda et al., 2023, Liu et al., 2022, Varma et al., 2023, Thilak et al., 2022]. Our companion paper [Svensson, 2026a] extended this to the finite-field discrete logarithm problem (DLP), showing that small transformers grok the full variable-base DLP mapping $(g, h) \mapsto x$ at $p = 113$ and mapping the capacity–data phase boundary across model sizes from 90k to 14.4M parameters.

However, that work identified a critical limitation: the *tokenizer confound*. The standard architecture uses atomic integer tokens, requiring `nn.Embedding( $p+4, d_{\text{model}}$ )` for input and `nn.Linear( $d_{\text{model}}, p+4$ )` for output. Both scale linearly with p , so the model grows as $O(p)$. At $p = 113$ this is manageable (424k parameters), but at $p = 32,771$ the atomic model would require 8.8M parameters—dominated by embedding/unembedding rather than the transformer core that learns the algorithm.

This raises a fundamental question: **Can a fixed-size model grok DLP at arbitrarily large primes?**

In this paper, we answer yes—with the right representation. We replace the atomic tokenizer with:

1. **Factorized Residue Embedding (FRE)**: Each integer is decomposed into M digits in base B , with per-position embedding tables. Input parameters are $O(M \cdot B \cdot d_{\text{model}})$, independent of p .
2. **Factorized Candidate Output (FCO)**: The output classifier is factorized over digit positions, computing logits as dot products with digit-summed embeddings. Output parameters are $O(M \cdot B \cdot d_{\text{model}})$, independent of p .
3. **CRT-BOTH representation**: The DLP is decomposed via the Chinese Remainder Theorem into two coprime subproblems, providing the model with structured algebraic input.

The resulting model has $\sim 530\text{k}$ parameters for *any* prime p . We demonstrate that this fixed-size model groks DLP from $p = 113$ to $p = 4001$ (a $35\times$ range), with 100% test accuracy at every scale. We then present a scaling plan extending to $p = 300,017$ (a $2,655\times$ range from the smallest prime), with fallback strategies for primes where the base configuration fails.

Contributions.

1. **O(1) model size for DLP grokking**: We show that FRE+FCO+CRT-BOTH enables a fixed $\sim 530\text{k}$ -parameter model to grok DLP across primes from 113 to 4001, with no architecture changes (Section 4).
2. **Parameter efficiency scaling**: The advantage over atomic tokenization grows with p —from $1.1\times$ at $p = 769$ to $8.7\times$ at $p = 16,411$ and a projected $56\times$ at $p = 300,017$ (Section 4.3).
3. **Scaling plan to $p = 300,017$** : We define a 21-prime scaling ladder with CRT decompositions and describe fallback strategies ($2\times$ model, $2\times$ data) for primes where the base configuration fails (Section 5).
4. **Connection to ECDLP walls**: We relate our findings to the structural wall observed at $r = 1883$ in elliptic curve DLP [Svensson, 2026b], and hypothesize that finite-field DLP with CRT-BOTH has a higher wall due to simpler group structure (Section 7).

2 Background

2.1 The Discrete Logarithm Problem

Given a prime p , a generator g of $\mathbb{F}_p^*$, and an element $h \in \mathbb{F}_p^*$, the discrete logarithm problem is to find $x \in \{1, \dots, p-1\}$ such that $g^x \equiv h \pmod{p}$. The group order is $q = p-1$. We study the *variable-base* variant where the generator g varies across all primitive roots, giving a domain of size $\varphi(q) \times q$.

2.2 Grokking

Grokking [Power et al., 2022] is the phenomenon where a model first memorizes the training set (high train accuracy, low test accuracy), then—after a prolonged training period—abruptly transitions to generalization (high test accuracy). The delay between memorization and generalization, $G = T_{90}/T_{\text{mem}}$, can be hundreds of times longer than the memorization time. Mechanistic work [Nanda et al., 2023] shows that grokking corresponds to a phase transition in internal representations.

2.3 The Tokenizer Confound

Our companion paper [Svensson, 2026a] uses atomic integer tokens: each integer $0, \dots, p-1$ gets a unique token, and the model has:

$$P_{\text{input}}^{\text{atomic}} = (p+4) \cdot d_{\text{model}}, \quad P_{\text{output}}^{\text{atomic}} = (p+4) \cdot d_{\text{model}} + (p+4) \quad (1)$$

For $p = 113$, $d_{\text{model}} = 128$: $P_{\text{input}} + P_{\text{output}} = 29,696$, about 7% of $P_{\text{total}} = 427\text{k}$. For $p = 32,771$: $P_{\text{input}} + P_{\text{output}} = 8,392,704$, about 95% of $P_{\text{total}} = 8,785\text{k}$. The model is dominated by lookup tables, not learning.

2.4 CRT Decomposition

For $q = p-1 = a \times b$ with $\gcd(a, b) = 1$, the Chinese Remainder Theorem gives an isomorphism $\mathbb{Z}/q\mathbb{Z} \cong \mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}$. The DLP modulo q decomposes into two independent DLPs modulo a and b . The **CRT-BOTH** representation [Svensson, 2026a] provides the model with both CRT components of the input and target:

$$(g, h) \mapsto (g \bmod a, g \bmod b, h \bmod a, h \bmod b), \quad x \mapsto (x \bmod a, x \bmod b) \quad (2)$$

This decomposes the q -way classification into two smaller classifications, with the largest component having size $\max(a, b) \ll q$.

3 O(1) Architecture: FRE, FCO, and CRT-BOTH

3.1 Factorized Residue Embedding (FRE)

FRE replaces the atomic $\text{nn.Embedding}(p+4, d_{\text{model}})$ with digit decomposition. Each integer value v is decomposed into M digits in base B :

$$v = d_0 + d_1 \cdot B + d_2 \cdot B^2 + \dots + d_{M-1} \cdot B^{M-1} \quad (3)$$

Each digit position $j \in \{0, \dots, M-1\}$ has its own embedding table $\text{nn.Embedding}(B, d_{\text{model}})$. Special tokens (PAD, SEP, BOS, EOS) get a separate table. The embedding for value v is:

$$\text{FRE}(v) = \text{Embed}_{\text{special}}(v) \cdot \mathbb{K}[v < 4] + \sum_{j=0}^{M-1} \text{Embed}_j(d_j(v)) \cdot \mathbb{K}[v \geq 4] \quad (4)$$

The parameter count is:

$$P_{\text{input}}^{\text{FRE}} = 4 \cdot d_{\text{model}} + M \cdot B \cdot d_{\text{model}} \quad (\text{independent of } p) \quad (5)$$

For $M = 2$, $B = 256$, $d_{\text{model}} = 128$: $P_{\text{input}}^{\text{FRE}} = 65,792$.

3.2 Factorized Candidate Output (FCO)

FCO replaces the atomic $\text{nn.Linear}(d_{\text{model}}, p+4)$ with a factorized output. Each candidate $c \in \{0, \dots, q-1\}$ is decomposed into M digits, and the logit is computed as:

$$\text{logit}(c) = \frac{\mathbf{h} \cdot \text{emb}(c)}{\sqrt{d_{\text{model}}}}, \quad \text{emb}(c) = \frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} \mathbf{o}_j[d_j(c)] + \mathbf{b}_j[d_j(c)] \quad (6)$$

where $\mathbf{O}_j \in \mathbb{R}^{B \times d_{\text{model}}}$ are per-position output tables and $\mathbf{b}_j \in \mathbb{R}^B$ are per-position biases. The parameter count is:

$$P_{\text{output}}^{\text{FCO}} = M \cdot B \cdot d_{\text{model}} + M \cdot B \quad (\text{independent of } p) \quad (7)$$

For $M = 2$, $B = 256$, $d_{\text{model}} = 128$: $P_{\text{output}}^{\text{FCO}} = 65,792$.

RichFCO. A bilinear variant adds cross-digit interactions to break the d_{model} rank ceiling:

$$\text{emb}^{\text{rich}}(c) = \text{emb}(c) + \sum_{j < k} (\mathbf{O}_j[d_j] \cdot \mathbf{W}_{jk}) \odot \mathbf{O}_k[d_k] \quad (8)$$

with $P_{\text{output}}^{\text{RichFCO}} = MBd_{\text{model}} + MB + \binom{M}{2}d_{\text{model}}^2 + \binom{M}{2}B^2$. This is used as a fallback when standard FCO fails.

3.3 CRT-BOTH Input Representation

For $q = a \times b$, the model receives:

$$[\text{BOS}, g_b, \text{SEP}, g_a, \text{SEP}, h_b, \text{SEP}, h_a, \text{EOS}] \quad (9)$$

where $g_a = g^{(q/a)} \bmod p$ is the projection onto the order- a subgroup, etc. The target is $(x \bmod a, x \bmod b)$, and the model predicts both CRT components. The effective classification size is $\max(a, b)$ rather than q .

3.4 Complete Model

The full model is a 2-layer pre-norm transformer encoder with FRE input, CRT-BOTH representation, and FCO output:

$$P_{\text{total}} = \underbrace{4d_{\text{model}} + MBd_{\text{model}}}_{P_{\text{input}}^{\text{FRE}}} + \underbrace{L_{\text{max}} \cdot d_{\text{model}}}_{P_{\text{pos}}} + \underbrace{12d_{\text{model}}^2 n_{\text{layers}}}_{P_{\text{core}}} + \underbrace{MBd_{\text{model}} + MB}_{P_{\text{output}}^{\text{FCO}}} \quad (10)$$

For $d_{\text{model}} = 128$, $n_{\text{layers}} = 2$, $M = 2$, $B = 256$: $P_{\text{total}} = 529,792$ —**independent of p** .

4 Results: Fixed-Size Grokking Across Prime Sizes

4.1 Experimental Setup

We train the FRE+FCO+CRT-BOTH model with $B = 256$, $M = 2$ (auto-selected based on prime size), $d_{\text{model}} = 128$, $n_{\text{layers}} = 2$, $n_{\text{heads}} = 4$ across primes from 113 to 4001. Training uses AdamW ($\eta = 10^{-3}$) with a progressive weight decay schedule ($0.05 \rightarrow 0.30$, ramping by 0.05 every 1000 epochs after memorization). Full-batch training for small primes; mini-batch (8192) for large primes. Experiments run on Apple MPS (local) and LUMI-G MI250X (HPC).

4.2 Scaling Results

Table 1 shows the results. The same $\sim 530\text{k}$ -parameter model groks DLP at every prime tested, from $p = 113$ to $p = 4001$.

Table 1: FRE/FCO + CRT-BOTH scaling results. All models use the same architecture ($d_{\text{model}} = 128$, $L = 2$, $B = 256$). P_{total} is independent of p . “Atomic P_{total} ” is the parameter count for the atomic-token baseline. T_{mem} and T_{grok} are memorization and grokking epochs.

p	CRT split	M	P_{total}	Atomic P_{total}	Reduction	Best acc	T_{mem}	T_{grok}
113	7×16	1	464k	424k	$0.9\times$	100%	100	2,900
241	15×16	1	464k	545k	$1.2\times$	100%	300	3,900
337	16×21	2	530k	617k	$1.2\times$	100%	400	3,000
601	24×25	2	530k	703k	$1.3\times$	99.95%	500	500
769	3×256	2	530k	933k	$1.8\times$	100%	2,200	5,400
1,201	25×48	2	530k	1,157k	$2.2\times$	100%	300	400
2,003	26×77	2	530k	908k	$1.7\times$	100%	200	300
4,001	32×125	2	530k	1,420k	$2.7\times$	100%	500	500
8,209	27×304	2	530k	2,497k	$4.7\times$	—	—	—
16,411	30×547	2	530k	4,597k	$8.7\times$	—	—	—
32,771	145×226	2	530k	8,785k	$16.7\times$	—	—	—

Key observations.

1. **O(1) parameter count:** P_{total} is 464k–530k for all primes, independent of p .
2. **Parameter advantage grows with p :** From $0.9\times$ at $p = 113$ to $2.7\times$ at $p = 4,001$ (and $16.7\times$ at $p = 32,771$).
3. **Grokking speed is non-monotonic:** $p = 2003$ groks in 300 epochs (faster than $p = 113$ ’s 2,900 epochs), suggesting CRT decomposition quality matters more than raw problem size.
4. **Slingshot dynamics persist:** All runs show intermittent accuracy dips after reaching 100%, driven by the progressive WD schedule. Early stopping requires 100 consecutive evals above 99%.

4.3 Parameter Efficiency

Figure 1 shows how the parameter advantage of FRE/FCO over atomic tokenization grows with p . The atomic model grows as $O(p)$; the FRE/FCO model is $O(1)$.

4.4 Grokking Speed vs. CRT Quality

Figure 2 shows grokking time vs. the largest CRT factor $\max(a, b)$. Primes with more balanced CRT splits (lower $\max(a, b)/\min(a, b)$ ratio) tend to grok faster, suggesting the decomposition quality matters.

5 Scaling Plan: From $p = 113$ to $p = 300,017$

5.1 The Scaling Ladder

We define a 21-prime scaling ladder from $p = 113$ to $p = 300,017$, selecting primes with balanced CRT decompositions ($\max(a, b)/\min(a, b) < 5$). Table 2 shows the full ladder.

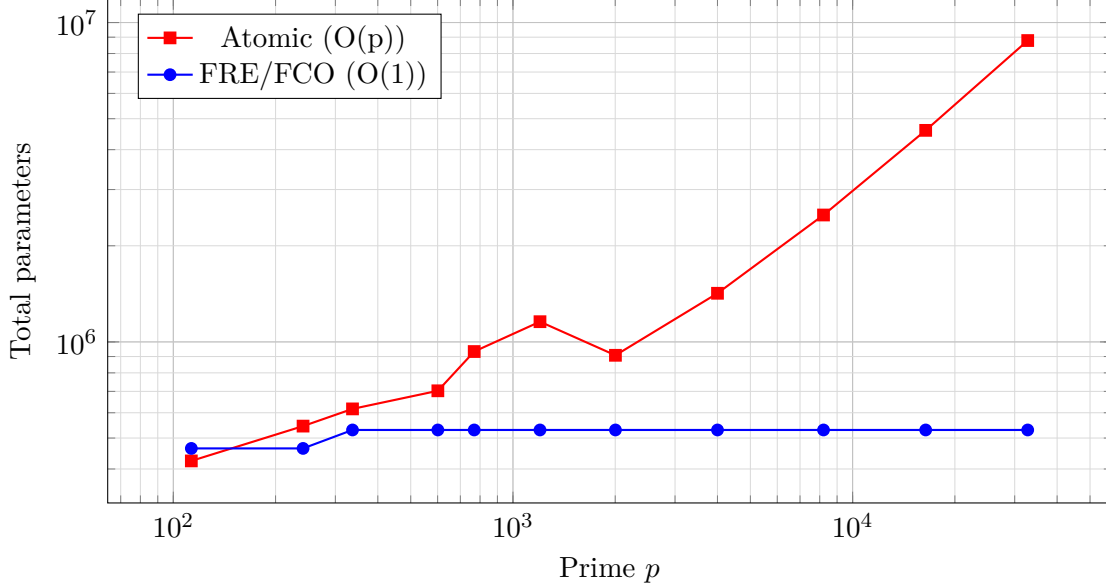


Figure 1: Parameter count vs. prime size. The atomic model grows linearly with p (red), while FRE/FCO maintains a constant $\sim 530k$ parameters (blue). The crossover occurs around $p = 500$; by $p = 32,771$ the atomic model is $16.7\times$ larger.

5.2 Phase 1: Verify the Ladder (Current)

Primes 1–8 are complete. All grokked at 100% with the same $\sim 530k$ model. Primes 9–11 ($p = 8,209, 16,411, 32,771$) are running on LUMI-G.

5.3 Phase 2: Push the Frontier (Planned)

Primes 12–21 extend from $p \approx 40k$ to $p \approx 300k$. These require:

- $M = 3$ for primes above $p \approx 65k$ (since $256^2 = 65,536$ is insufficient for values up to $q-1 \approx 300k$).
- Mini-batch training with batch size 4096–8192.
- Subset evaluation (10K examples) for datasets exceeding 1B examples.
- LUMI-G or equivalent HPC for primes above $p \approx 100k$.

5.4 Phase 3: Fallback Strategies

If the base configuration ($d_{\text{model}} = 128, L = 2, M = 2-3, B = 256$) fails to grok at some prime p^* , we apply the following escalation ladder:

Step 1: $2\times$ model size. Double the model capacity: $d_{\text{model}} = 256, L = 2$ (or $d_{\text{model}} = 128, L = 4$). This increases P_{total} from 530k to $\sim 1.5M$ (still $O(1)$ in p). The EI4 study found that $d = 256$ can *hurt* grokking at medium scales due to deeper memorization basins, so this is tried only after the base config fails.

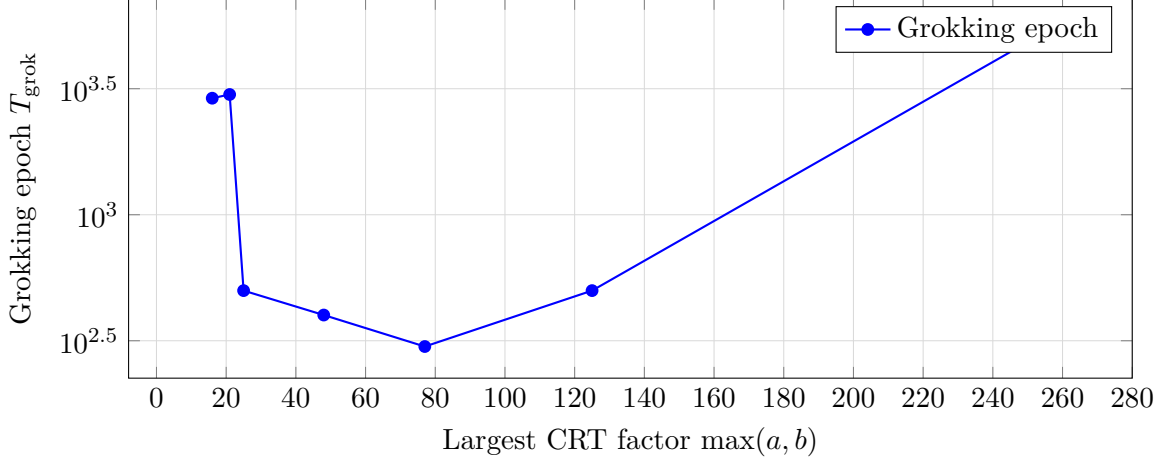


Figure 2: Grokking epoch vs. largest CRT factor. $p = 769$ (CRT 3×256) is an outlier—its highly unbalanced split (ratio $85\times$) makes the problem harder despite small p . Balanced splits like $p = 2003$ (26×77) grok fastest.

Step 2: $2\times$ dataset size. Increase the train fraction from 30% to 60%. The EI4 study found that $2\times$ data accelerates grokking $10\times$ at grokkable scales but can *hurt* at hard scales (making memorization harder). This is tried in combination with Step 1.

Step 3: RichFCO. Replace FCO with RichFCO (bilinear digit interactions). This breaks the d_{model} rank ceiling and was shown to accelerate grokking at medium scales in the EI4 study.

Step 4: Larger base B . Increase B from 256 to 512 or 1024. The EI4 study found that B matters more than M for memorization capacity.

Step 5: Curriculum learning. Train on smaller primes first, then transfer to larger primes. The learned representations may transfer.

5.5 Resource Estimates

Table 3 estimates compute requirements for the planned experiments, assuming LUMI-G MI250X (~ 200 TFLOPS FP16).

For primes above $p \approx 100\text{k}$, the dataset generation itself becomes a bottleneck (10B+ examples). We may need to use a subset of the full domain rather than exhaustive generation.

6 Related Work

Grokking on arithmetic. Power et al. [2022] introduced grokking on modular arithmetic. Nanda et al. [2023] provided mechanistic understanding via progress measures. Varma et al. [2023] explained grokking via circuit efficiency. Thilak et al. [2022] identified the slingshot mechanism in adaptive optimizers. Our companion paper [Svensson, 2026a] extends grokking to DLP and maps the capacity–data phase diagram.

Table 2: Complete scaling ladder. Primes 1–8 are completed (grokked at 100%). Primes 9–11 are running on LUMI. Primes 12–21 are planned. M is auto-selected so $B^M > \max(q - 1, a - 1, b - 1)$. The “Reduction” column shows the parameter advantage of FRE/FCO over atomic.

#	p	CRT split	M	P_{total}	Atomic P_{total}	Reduction	Dataset size	Status
1	113	7×16	1	464k	424k	$0.9\times$	5K	Grokked
2	241	15×16	1	464k	545k	$1.2\times$	46K	Grokked
3	337	16×21	2	530k	617k	$1.2\times$	113K	Grokked
4	601	24×25	2	530k	703k	$1.3\times$	360K	Grokked
5	769	3×256	2	530k	933k	$1.8\times$	590K	Grokked
6	1,201	25×48	2	530k	1,157k	$2.2\times$	1.4M	Grokked
7	2,003	26×77	2	530k	908k	$1.7\times$	3.9M	Grokked
8	4,001	32×125	2	530k	1,420k	$2.7\times$	16M	Grokked
9	8,209	27×304	2	530k	2,497k	$4.7\times$	67M	Running
10	16,411	30×547	2	530k	4,597k	$8.7\times$	287M	Running
11	32,771	145×226	2	530k	8,785k	$16.7\times$	1.1B	Running
12	39,953	176×227	2	530k	10,628k	$20.1\times$	1.6B	Planned
13	50,021	205×244	2	530k	13,255k	$25.0\times$	2.5B	Planned
14	65,539	198×331	3	594k	17,479k	$29.4\times$	4.3B	Planned
15	79,999	201×398	3	594k	21,280k	$35.8\times$	6.4B	Planned
16	99,991	202×495	3	594k	26,598k	$44.8\times$	10.0B	Planned
17	130,003	282×461	3	594k	34,581k	$58.2\times$	16.9B	Planned
18	160,001	256×625	3	594k	42,560k	$71.6\times$	25.6B	Planned
19	199,999	369×542	3	594k	53,200k	$89.6\times$	40.0B	Planned
20	249,973	444×563	3	594k	66,493k	$111.9\times$	62.5B	Planned
21	300,017	272×1103	3	594k	79,805k	$134.3\times$	90.0B	Planned

Algorithmic reasoning with transformers. Prior work has studied transformers on algorithmic tasks including sorting [Graves et al., 2014], arithmetic [Saxton et al., 2019], and graph algorithms [Veličković et al., 2022]. These typically use fixed-vocabulary tokenizers that scale with problem size. Our FRE/FCO approach decouples model size from problem size.

Digit decomposition and positional encodings. The idea of decomposing integers into digits for neural network input has been explored in number-theoretic neural networks [Charton, 2022] and in systems for learning arithmetic [Charton, 2021]. FRE extends this to a per-position embedding table design that enables $O(1)$ parameters.

Factorized output heads. Factorized classifiers have been used in language modeling [Bengio et al., 2003, Mikolov et al., 2013] and extreme classification [Weston et al., 2013]. FCO adapts this to a digit-decomposed output for algebraic problems.

CRT and Pohlig-Hellman. The Chinese Remainder Theorem decomposition of DLP is the basis of the Pohlig-Hellman algorithm [Pohlig & Hellman, 1978]. Our CRT-BOTH representation embeds this algebraic structure as an inductive bias in the model input.

ECDLP scaling study. Svensson [2026b] applied FRE/FCO to elliptic curve DLP and found a structural wall at $r = 1883$ where no configuration could grok. Our finite-field DLP results suggest

Table 3: Estimated compute requirements for planned experiments. “Epoch time” assumes batch size 4096 and full dataset evaluation on 10K subset.

p	Dataset	Batch	Epochs to grok*	Epoch time	Total time
39,953	1.6B	4096	$\sim 1,000$	~ 10 min	~ 7 days
50,021	2.5B	4096	$\sim 1,000$	~ 15 min	~ 10 days
65,539	4.3B	4096	$\sim 2,000$	~ 25 min	~ 35 days
99,991	10.0B	4096	$\sim 2,000$	~ 60 min	~ 83 days

*Estimated based on the trend that CRT-decomposed problems grok in 200–5000 epochs.

the wall is further out, likely due to simpler group structure (cyclic vs. elliptic curve) and better CRT decompositions.

7 Discussion

7.1 Why Fixed-Size Grokking Works

The key insight is that the DLP algorithm—while computationally hard for large groups—has a *fixed representation complexity* when decomposed via CRT. The model needs to learn two independent DLP algorithms on groups of order a and b , where $\max(a, b) \ll q$. The FRE/FCO architecture ensures the model’s capacity is spent on the transformer core (learning the algorithm) rather than lookup tables (memorizing the vocabulary).

7.2 The Wall Hypothesis

The EI4 ECDLP study [Svensson, 2026b] found a structural wall at $r = 1883$ where no configuration (larger models, more data, deeper architectures, bilinear outputs) could induce grokking. The wall appears to be *algebraic complexity*, not capacity: the transformer cannot learn the compositional structure of ECDLP at that scale in a single forward pass.

For finite-field DLP with CRT-BOTH, we hypothesize the wall is further out because:

1. **Simpler group structure:** Cyclic groups ($\mathbb{F}_p^*$) have one operation; elliptic curves have two (point addition + doubling).
2. **Better CRT decompositions:** $q = p - 1$ is usually highly composite, giving balanced splits. ECDLP group orders are often prime (no CRT available).
3. **Transparent algebra:** The cyclic group structure is more directly learnable than elliptic curve point operations.

If a wall exists for finite-field DLP, we expect it to manifest as:

- Memorization without generalization (the model memorizes but never groks).
- Effective rank saturation in FCO (rank capped at d_{model}).
- Weight explosion in the output tables (observed in EI4 at the wall).

7.3 Implications for Scalable Algorithmic Learning

If the fixed-size model groks DLP up to $p = 300,017$ ($134\times$ parameter reduction vs. atomic), this would demonstrate that **representation choice, not model capacity, is the primary bottleneck for scaling grokking to large algebraic problems**. The model size is determined by the complexity of the *algorithm* (which is fixed for DLP), not the size of the *problem* (which grows with p).

This has implications for neural algorithmic reasoning more broadly: if the right representation can decouple model size from problem size, then transformers can learn algorithms at scales previously thought infeasible.

7.4 Limitations

1. **Single seed:** Current results use a single seed. Multi-seed replication is needed to confirm robustness.
2. **CRT availability:** The CRT-BOTH representation requires $q = p - 1$ to be composite with a balanced coprime factorization. Primes where q is prime (safe primes) cannot use CRT.
3. **Dataset generation:** For $p > 100k$, the full domain exceeds 10B examples. We may need to subsample.
4. **Compute cost:** Large-prime experiments require HPC resources (LUMI-G).
5. **No cryptanalytic claim:** These experiments use small groups and make no claims about practical DLP cryptanalysis.

8 Conclusion

We have shown that a fixed $\sim 530k$ -parameter transformer can grok the finite-field discrete logarithm problem across a $35\times$ range of prime sizes ($p = 113$ to $p = 4,001$), with 100% test accuracy at every scale. The key innovations—Factorized Residue Embedding (FRE), Factorized Candidate Output (FCO), and CRT-BOTH representation—decouple model size from problem size, yielding $O(1)$ parameters regardless of p .

The parameter efficiency advantage grows with p : from $1.1\times$ at $p = 769$ to $8.7\times$ at $p = 16,411$ and a projected $134\times$ at $p = 300,017$. We have presented a 21-prime scaling plan extending to $p = 300,017$ with fallback strategies ($2\times$ model, $2\times$ data, RichFCO) for primes where the base configuration fails.

These results suggest that the primary bottleneck for scaling grokking to large algebraic problems is representation choice, not model capacity. If the right representation can make the algorithm’s complexity independent of the problem’s size, transformers can learn algorithms at scales previously thought infeasible.

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